Software Test Plan (STP) - Task Manager Application

Project: Task Manager Application
Version: 1.0
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1. Introduction

Purpose: This document defines the test plan for the Task Manager Application v1.0. It outlines objectives, scope, strategy, resources, schedule and responsibilities for testing.

Scope: Testing covers authentication, task management, projects and organisation, search/filter, reminders, Kanban board, dashboard, activity history, comments, CSV export and administration, together with performance, security, accessibility and browser-compatibility checks. Third-party services (SMTP provider, cloud hosting internals) are excluded.

References: Task Manager SRS v1.0, Task Manager SAD v1.0, OWASP ASVS / Top 10, WCAG 2.1.

Definitions: SRS (Software Requirements Specification), SAD (Software Architecture and Design), RTM (Requirements Traceability Matrix), UAT (User Acceptance Testing), JWT, RBAC, API, UI.

2. Test Items

- Authentication module (register, login, lock-out, password reset)
- Task management module (create, edit, delete/restore, status, assignment)
- Project and organisation module (projects, members, categories, tags)
- Search / filter / sort
- Notification and reminder module
- Kanban board and progress dashboard
- Activity history and comments
- Report / CSV export module
- Admin console (user and role management)
- REST API and database layer

3. Features to be Tested

Features mapped to SRS requirement IDs:

Req ID	Feature	Priority
TM-F-001	Register account	High
TM-F-002	Login / logout	High
TM-F-003	Account lock-out	High
TM-F-004	Password reset	Medium
TM-F-010	Create task	High
TM-F-011	Edit task	High
TM-F-012	Delete / restore task	Medium
TM-F-013	Task status	High

Req ID	Feature	Priority
TM-F-014	Assign task	High
TM-F-020	Manage projects	High
TM-F-021	Category & tags	Medium
TM-F-022	Search & filter	High
TM-F-023	Reminders	Medium
TM-F-030	Kanban board	High
TM-F-031	Progress dashboard	High
TM-F-032	Activity history	Medium
TM-F-040	Task comments	Medium
TM-F-041	User administration	High
TM-F-042	CSV export	Low
TM-F-043	Project access control	High
TM-NF-001	Response time	High
TM-NF-002	Availability	High
TM-NF-003	Data protection	High
TM-NF-004	Accessibility / responsive	Medium
TM-NF-005	Scalability	Medium
TM-NF-006	Browser support	Medium
TM-NF-007	Backup & recovery	Medium
TM-SR-001	TLS 1.2+	High
TM-SR-002	Password hashing	High
TM-SR-003	Token expiry	High
TM-SR-004	RBAC enforcement	High
TM-SR-005	Input validation	High
TM-SR-006	Rate limiting	Medium
TM-SR-007	Security logging	Medium

4. Features Not to be Tested

- Internal behaviour of third-party libraries and frameworks (assumed tested by their vendors)
- SMTP provider delivery infrastructure (only hand-off to the provider is tested)
- Cloud hosting platform internals and network hardware
- Native mobile apps, file attachments and calendar integration (not in v1.0 scope)

5. Test Approach / Strategy

Levels:

- Unit tests (functions, services, components) - developers, Jest
- Integration tests (UI-API, API-database, API-SMTP stub)
- System tests (end-to-end user flows in a production-like environment)
- Acceptance tests (UAT with product owner / sample users)

Types:

- Functional testing (all FRs)
- Regression testing (automated suite on every build)

- Performance testing (response time, load, scalability)
- Usability and accessibility testing
- Compatibility testing (Chrome, Firefox, Edge, Safari)
- Recovery testing (backup and restore)

Entry Criteria: Stable build deployed to the test environment, smoke test passed, test data available, test cases reviewed.

Exit Criteria: 100% of planned test cases executed, 95% pass rate with no open critical or high-severity defects, all High-priority requirements verified and all acceptance criteria satisfied.

5.1 Security Validation

- Verify password hashing (bcrypt) and absence of plaintext passwords/tokens in DB and logs
- TLS 1.2+ verification with an SSL scanner; HTTP-to-HTTPS redirect
- JWT expiry, revocation on logout and tampered-token rejection
- RBAC / broken-access-control tests: member vs admin, project membership (IDOR checks)
- Automated OWASP ZAP scan and manual tests for SQL injection and XSS
- Rate-limit and brute-force lock-out tests on login and reset endpoints
- Fuzzing of input fields (title, description, search, comment)

6. Test Environment

Item	Details
Hardware	Test server / cloud VM (2 vCPU, 4 GB RAM); client PCs, one tablet and one smartphone
Software	Task Manager web app v1.0 (React + Node.js/Express), PostgreSQL 15, SMTP sandbox (e.g. MailHog)
Browsers	Latest two versions of Chrome, Firefox, Edge; Safari (macOS/iOS)
Tools	Jest (unit), Cypress / Selenium (UI automation), Postman / Newman (API), JMeter (performance), OWASP ZAP (security), Lighthouse / axe (accessibility), Jira or GitHub Issues (defect tracking)
Test Data	Dummy users (Admin, Project Owner, Members), sample projects and 10,000+ generated tasks for load tests

7. Test Schedule

Milestones:

Milestone	Date
Test plan approved / test case design starts	05-Oct-2026
Test case design complete	09-Oct-2026
Environment setup and smoke test	12-Oct-2026
Test execution start (cycle 1)	13-Oct-2026
Regression cycle (cycle 2)	26-Oct-2026 to 30-Oct-2026
Test execution end	30-Oct-2026
UAT	02-Nov-2026 to 06-Nov-2026
Test Summary Report	09-Nov-2026

8. Test Deliverables

- Test Plan (this document)

- Test Cases (manual and automated)
- Test Scripts (Cypress / Postman / JMeter)
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	<Name>	Prepare plan, coordinate execution, report status
Test Engineer	<Name>	Design and execute test cases, automate regression, log defects
Developer	<Name>	Unit tests, fix defects, support triage
Product Owner	<Name>	Approve test results, take part in UAT, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds; daily CI builds
Test environment downtime	Keep a backup environment (Docker Compose / cloud VM)
Email / SMTP sandbox unavailable	Use local mail stub (MailHog); test hand-off only
Insufficient time for performance and security testing	Automate early; start scans as soon as the API is stable
Requirement changes late in the cycle	Change control via RTM; re-plan affected test cases

11. Assumptions & Dependencies

- A stable test environment with realistic data volume will be available before execution.
- Test accounts and seed data will be provided by developers.
- SMTP sandbox and CI pipeline are set up before cycle 1.
- SRS v1.0 requirements are baselined and changes follow change control.

12. Suspension & Resumption Criteria

Suspend testing if:

- Environment unavailable for more than 4 hours
- Build is so unstable that it blocks more than 30% of test cases
- A critical defect prevents login or data access for all users

Resume testing if:

- Blocking defects are fixed and verified with a smoke test
- Environment is stabilised

13. Test Case Management & Traceability

The RTM (SRS Section 8) maps every requirement to its test cases; the mapping below is maintained by the QA Lead and updated as test cases are added.

Requirement	Description	Test case(s)
TM-F-001	Register account	TC-Auth-01
TM-F-002	Login / logout	TC-Auth-02
TM-F-003	Account lock-out	TC-Auth-03
TM-F-004	Password reset	TC-Auth-04
TM-F-010	Create task	TC-Task-01
TM-F-011	Edit task	TC-Task-02
TM-F-012	Delete / restore task	TC-Task-03
TM-F-013	Task status	TC-Task-04
TM-F-014	Assign task	TC-Task-05
TM-F-020	Manage projects	TC-Proj-01
TM-F-021	Category & tags	TC-Proj-02
TM-F-022	Search & filter	TC-Srch-01
TM-F-023	Reminders	TC-Rem-01
TM-F-030	Kanban board	TC-Brd-01
TM-F-031	Progress dashboard	TC-Dash-01
TM-F-032	Activity history	TC-Hist-01
TM-F-040	Task comments	TC-Cmt-01
TM-F-041	User administration	TC-Adm-01
TM-F-042	CSV export	TC-Exp-01
TM-F-043	Project access control	TC-Adm-02
TM-NF-001	Response time	TC-Perf-01
TM-NF-002	Availability	Ops report
TM-NF-003	Data protection	TC-Sec-02
TM-NF-004	Accessibility / responsive	TC-UX-01
TM-NF-005	Scalability	TC-Perf-02
TM-NF-006	Browser support	TC-Comp-01
TM-NF-007	Backup & recovery	TC-OPS-01
TM-SR-001	TLS 1.2+	TC-Sec-01
TM-SR-002	Password hashing	TC-Sec-02
TM-SR-003	Token expiry	TC-Sec-03
TM-SR-004	RBAC enforcement	TC-Sec-04
TM-SR-005	Input validation	TC-Sec-05
TM-SR-006	Rate limiting	TC-Sec-06
TM-SR-007	Security logging	TC-Sec-07

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed / failed / blocked

- Defect density (defects per module) and defect severity distribution
- Defect aging
- Requirement coverage (via RTM)
- Automation coverage

Reports: daily execution status; weekly summary; final Test Summary Report.

15. Approvals

Role	Name	Signature / Date
QA Lead		
Dev Lead		
Product Owner		

Appendix A - Sample Detailed Test Cases

Test case	Steps (summary)	Expected result	Req
TC-Auth-01	Open Register page; enter valid name, unique email, strong password; submit	Account created; confirmation shown. Same email again gives duplicate-email error	TM-F-001
TC-Auth-03	Enter wrong password 5 times for one user	5th attempt locks account 15 min; audit event recorded; correct password rejected while locked	TM-F-003
TC-Task-01	Click New Task; fill title, due date, priority High; save. Repeat with empty title	Task appears in list with correct values; empty title rejected with message	TM-F-010
TC-Task-04	Change status To Do -> In Progress -> Done	Status updates in list and board; completed timestamp stored	TM-F-013
TC-Srch-01	Filter by status Done, priority High and keyword 'report'	Only tasks matching all filters shown in under 2 s	TM-F-022
TC-Rem-01	Create task due in 24 h; run scheduler	In-app and email reminder delivered; overdue task marked red after due date	TM-F-023
TC-Brd-01	Drag card from To Do to Done	Card moves and status persists after refresh	TM-F-030
TC-Adm-02	Member A requests project data of a project they don't belong to (API)	HTTP 403 returned; no data leaked	TM-F-043
TC-Perf-01	JMeter: 500 concurrent users, mixed workload, 15 min	95th percentile page < 2 s, API < 500 ms; error rate < 1%	TM-NF-001
TC-Sec-05	Run ZAP scan; submit SQL / script payloads in text inputs	No injection or XSS findings of High severity	TM-SR-005
TC-UX-01	Run Lighthouse / axe; navigate app by keyboard only	No critical accessibility violations; all functions reachable	TM-NF-004